

B.Tech III Year II Semester (R19) Regular Examinations July/August 2022

INTRODUCTION TO MACHINE LEARNING

(Common to CSE & IT)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- (a) List the basic design issues to machine learning. 2M
 - (b) State version space representation theorem. 2M
 - (c) What is the representational power of perceptrons? 2M
 - (d) How to compute expected value and variance of random variable? 2M
 - (e) State bayes theorem. 2M
 - (f) Under what conditions is successful learning possible? 2M
 - (g) How to use entropy as evaluation function? 2M
 - (h) What factors contribute to the popularity of genetic algorithm? 2M
 - (i) What is the essential difference between analytical and inductive learning? 2M
 - (j) What are the limitations of explanation-based learning? 2M

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Explain the concept of inductive bias in brief. 5M
- (b) Describe hypothesis space search in decision tree learning. 5M

OR

- 3 (a) How the concept learning can be viewed as the task of searching and explain? 5M
- (b) Consider the given below training example which finds malignant tumors from MRI scans. 5M

Example	Shape	Size	Color	Surface	Thickness	Target concept
1	Circular	Large	Light	Smooth	Thick	Malignant
2	Circular	Large	Light	Irregular	Thick	Malignant
3	Oval	Large	Dark	Smooth	Thin	Benign
4	Oval	Large	Light	Irregular	Thick	Malignant
5	Circular	Small	Light	Smooth	Thick	Benign

- 4 (a) List the appropriate problems for neural network learning. 5M
- (b) Discuss the perceptron training rule and delta rule that solves the learning problem of perceptron. 5M

OR

- 5 (a) Write a remark on representation of feed forward networks. 5M
- (b) Explain about Lamarckian evolution. 5M

- 6 (a) Explain maximum likelihood and least squared error hypothesis. 5M
- (b) Consider a football game between two rival teams: Team 0 and Team 1. Suppose Team 0 wins 95% of the times and Team 1 wins the remaining matches. Among the games won by team 0, only 30% of them come from playing on teams 1's football field. On the other-hand 75% of the victories for team 1 are obtained while playing at home. If team 1 is to host the next match between the two teams, which team will most likely emerge as the winner? 5M

OR

Contd. in page 2

- 7 (a) Explain Bayesian belief networks. 5M
(b) Prove that how maximum likelihood (Bayesian learning) can be used in any learning algorithms that are used to minimize the squared error between actual output hypothesis and predicted output hypothesis. 5M
- 8 (a) Explain about simple error and true error. 5M
(b) Explain how reinforcement learning problem differs from other function approximation tasks. 5M
- OR**
- 9 (a) Define expected value, variance, standard deviation and estimate bias of a random variable. 5M
(b) Write a short note on: (i) Radial basis function. (ii) Locally weighted regression. 5M
(iii) Sampling theory.
- 10 (a) Write the Learn-One rule algorithm. 5M
(b) Discuss the inverse resolution rule for the first order logic. 5M
- OR**
- 11 (a) Justify whether Q-learning is a special case of temporal difference algorithm or not. 5M
(b) Explain sequential covering algorithm. 5M
